INTRODUCTION

Mosaic variegated aneuploidy research report

This document explains literature review, gene analysis, network analysis, pathway analysis, and their separate downstream drug-screening decisions.

> **Purpose.** Show what each analysis receives, what it does, which filters it uses, what it produces, and what its result can mean.

Research only | Not a diagnosis | No treatment recommendation | No patient-level fields shown

Each pipeline answers one question

Input

Material that enters a pipeline. Examples: article records, a gene list, or normalized identifiers.

Tool

Database or software that performs one step. Examples: PubMed, STRING, g:Profiler, or DAVID.

Filter

A rule that keeps or removes records. Filters reduce noise. They can also change results.

Output

A file or result from a pipeline. Outputs are evidence for follow-up. They are not final proof.

Terms used here. A *candidate gene* passes a research filter. A *network edge* is a reported protein association. An *enriched term* occurs more often than expected against a defined background.

METHOD WORKFLOW

The full analysis flow

Literature review
Collect and organize published evidence about genes, pathways, and reported biology.

→

Phase 0–1: gene analysis
Filter annotated VCF records. Convert passing records into candidate seed genes.

→

Phase 2: network pipeline
Check whether seed-gene proteins form known high-confidence associations.

→

Pathway pipeline
Test whether biological annotations occur more often than expected in selected genes.

Each stage changes the evidence type. A later stage does not prove an earlier-stage candidate is causal.

Pipeline	Question	Main output	Key limit
Literature review	What does published work mention and support?	Article and pathway summaries	Historical screen is not a complete systematic review.
Gene analysis	Which genes contain variants that meet research gates?	Candidate seed-gene list	Does not establish disease causality or allele phase.
Network pipeline	Which seed proteins have high-confidence STRING associations?	Edges and components	Does not show causal direction.
Pathway pipeline	Which annotations are over-represented?	Provider-specific term results	Does not show pathway activity.

Workflow sources: literature workflow. gene, network, and pathway workflow.

Literature review asks what published evidence already exists

Purpose

Find articles that mention named pathways or processes. Link articles to controlled pathway names and historical HPO terms.

Input

Eight HPO identifiers, 53 pathway synonym rules, exact PubMed queries, eligibility rules, and a frozen article snapshot.

Tools

PubMed ESearch and EFetch retrieve metadata. PMC resolves full-text availability. Reactome and KEGG support pathway identifiers.

Output

Search log, article registry, scored article table, pathway evidence table, rankings, heatmap, and UpSet summary.

Scope limit. This repository contains documentation of the literature workflow. The described patch and frozen scored article table are not installed here.

Source: literature workflow reproduction.

Article records move through retrieval, annotation, and review

Search
Run versioned Boolean queries. Record source, date, hit count, and query text.

→

Retrieve
Fetch metadata. Resolve available full text. Deduplicate records by PMID or DOI.

→

Annotate and map
Match configured pathway and HPO terms. Assign checked Reactome or KEGG references.

→

Filter and aggregate
Keep pathway mentions. Count distinct articles and HPO coverage by named pathway.

Important filters

- Pathway inclusion needs a literal pathway or configured process phrase.
- Gene-only mentions do not enter the pathway ranking.
- The old 48-paper manual subset is not an eligibility filter.

Required human review

- Screen titles, abstracts, and full text against prespecified rules.
- Record exclusions and reviewer disagreements.
- Use design-appropriate risk-of-bias assessment.

Source: literature workflow reproduction, stages 1–9.

Historical article filtering reduced 538 scored articles to 27 HPO-covered pathway groups

538
Scored articles in historical snapshot.

→

130
Articles with pathway or process mentions.

→

86
Articles with a pathway mention and any HPO or clinical mention.

→

27
Named pathway groups with at least one HPO feature covered.

What this result means

The screen finds literature-linked pathway groups. It helps decide which topics need closer review.

What this result does not mean

Article mentions do not prove a pathway is active, causal, or relevant in one person.

Conclusion. Historical counts describe an automated, reproducible corpus screen. They do not represent completed PRISMA screening, appraisal, or clinical evidence synthesis.

PHASE 0–1: GENE ANALYSIS

Gene analysis converts annotated VCF records into candidate seed genes

Input

18,252 annotated variant records from an upstream evidence table. Optional Exomiser files can add phenotype-ranking fields.

Tools

VEP and snpEff provide consequence annotation. gnomAD, ClinVar, CADD, SpliceAI, REVEL, and AlphaMissense supply supporting evidence.

Phase 0 output

A consolidated candidate-variant table. Each record stores gate status and a reason summary.

Phase 1 output

A gene mechanism matrix. A gene with one or more passing candidate variants becomes a network seed.

Plain-language purpose. This is a triage process. It keeps variants that are rare enough and damaging enough to justify research follow-up.

Source: JOURNEY.md, Phase 0 and Phase 1.

Three gate types determine whether a variant becomes a candidate

Check coordinates
Require a chromosome, positive position, reference allele, and alternate allele.

→

Check rarity
Keep low-frequency or gnomAD-not-found alleles. Threshold: allele frequency ≤ 0.001.

→

Check impact
Keep records that meet at least one functional or clinical evidence criterion.

→

Aggregate by gene
Genes with one or more passing records become Phase 1 seed genes.

Impact criteria
High-impact consequence, P/LP ClinVar, CADD ≥ 20, SpliceAI ≥ 0.2, REVEL ≥ 0.5, or AlphaMissense ≥ 0.5.

Special handling
Unknown frequency can pass only in an MVA-context gene and only with an impact pass. Genes with two or more passing records receive a multi-hit flag.

Code and report differ. Narrative reports list `NHOMALT < 5` as a rarity filter. Current code records population homozygotes but does not apply this threshold as a gate.

Gate implementation: candidate_consolidation.py. Reported thresholds: JOURNEY.md.

PHASE 0–1: GENE ANALYSIS RESULTS AND CONCLUSION

Gene analysis produced 1,985 filter-qualified candidate seed genes

18,252
Annotated variant records evaluated.

→

2,215
Variant records reported as passing research gates.

→

1,985
Genes with one or more passing candidate variants.

168
Genes with two or more passing candidate variants. This pattern needs phase confirmation.

6
Passing variants in canonical MVA genes, as reported in Phase 0.

Conclusion. The output is a filter-qualified candidate seed-gene list. It is useful input for network and pathway analysis. It is not a verified disease-gene list.

Reported counts: JOURNEY.md.

CANDIDATE-GENE RATIONALE

Candidate-gene rationale keeps established MVA biology separate from filter output

Published disease anchors

BUB1B, CEP57, TRIP13, CENATAC, MAD1L1, SLF2, and SMC5 have prior links to MVA-related chromosome-instability biology.

Research candidate gates

Rarity and impact filters select genes for follow-up. The gates do not reproduce published causality evidence.

Context checks

Network and pathway results add association context. They do not convert a candidate into a verified causal gene.

Why keep core genes

Published MVA genes remain biologically relevant even when they do not pass a case-specific variant gate. The later drug screen includes these core genes for this reason.

Why keep candidate labels

A passing record supports research priority only. Variant phase, functional effect, inheritance, and cellular rescue still need independent tests.

Interpretation rule. Do not call the 1,985 seed genes verified disease genes. Use published gene evidence and filter evidence as different evidence types.

Primary MVA mechanism: Hanks et al., 2004. Gene-context policy: JOURNEY.md, Phase 3.

PHASE 2: NETWORK PIPELINE

The network pipeline asks whether candidate seed genes show protein associations

Input

1,985 Phase 1 seed genes. Each gene has one or more passing candidate variants.

Tools

g:Profiler g:Convert standardizes symbols. STRING v12.5 provides protein association data.

Network filter

STRING combined score must be at least 0.700. The query adds zero extra proteins. Only seed-gene associations remain.

Output

Seed-gene list, STRING edge table, network summary, component summary, node table, and two HTML network views.

Normalize symbols
Map input symbols to official HGNC symbols with g:Convert.

→

Query STRING
Request associations at high confidence. Add no outside proteins.

→

Summarize graph
Count edges and components. Identify components that contain MVA-related genes.

Source: JOURNEY.md, Phase 2 parameters and outputs.

The network is sparse, with one reported MVA-focus component

1,985

Seed genes mapped to official HGNC symbols.

477

Reported STRING associations at score ≥ 0.700.

1,600

Reported connected components.

MVA-focus component

37 genes and 52 edges. Reported anchors include BUB1B, MAD1L1, and MAD2L2.

Why zero expansion matters

The graph contains no added neighbors. Therefore, every node began as a candidate seed gene.

Conclusion. The component supplies a focused set for follow-up. A STRING edge can describe physical binding, functional association, or co-annotation. It does not show a causal disease pathway.

PHASE 3: DRUG SCREENING DOWNSTREAM OF NETWORK PIPELINE

Network-based drug screening queries a defined 80-gene panel

Input
MVA-focus component, Phase 1 candidates, published core MVA genes, and quality-control targets.

→

Select panel
Apply symbol hygiene, core-gene inclusion, graph-neighbor, context-modifier, and control gates.

→

Query sources
Search Open Targets, ChEMBL, and DGIdb for drug-target evidence.

→

Output
Normalize source records into a drug-evidence matrix and a safety-ranking table.

7
Core MVA causal genes.

31
Graph-derived neighbors.

29
Context modifiers.

13
Drug-target controls.

Result. The 80-gene panel returned 2,212 source records and 2,006 unique drug-target hypotheses. This is a pharmacology evidence inventory, not a treatment list.

Reported Phase 3 methods and results: JOURNEY.md. Source roles: drug-repurposing context.

PHASE 3: SAFETY EVALUATION DOWNSTREAM OF NETWORK DRUG SCREENING

Safety evaluation rejects a drug action that can worsen a proposed defect

State defect direction
Record whether evidence supports loss, gain, abnormal dosage, or unknown function.

Check drug action and hazard
Compare inhibitor or activator action with the proposed direction. Record source evidence and pediatric hazard notes.

Assign research tier
Rank concordant hypotheses above uncertain records. Flag direct contradictions as hazards.

5
Concordant research hypotheses.

99
Off-target mitotic compensation hypotheses.

1,891
Hypotheses with uncertain direction.

11
Contraindicated hazards.

Safety result

15 records carry a mechanistic hazard flag. Thirty-eight of 80 targets have no retrieved drug evidence. Sparse direct druggability is an outcome, not a pipeline failure.

Safety limit

Database records cannot establish efficacy, exposure, or safety in MVA. None of these hypotheses is an approved MVA treatment.

Tier rules and reported results: JOURNEY.md, Phase 3. Evidence interpretation: drug-repurposing context.

PATHWAY PIPELINE: METHODS AND FILTERS

Pathway analysis tests annotations against a stated gene background

Select genes
Use candidate seed genes as the foreground. Choose a background gene universe.

→

Test annotations
Use Reactome, g:Profiler, or DAVID to test whether terms occur more often than expected.

→

Apply correction
Use the provider-adjusted value before interpreting an enrichment result.

Phase 2 pathway export

Reactome Analysis Service and g:Profiler use Phase 2 seed genes. The saved export has pathway names and FDR values.

Independent P0–P7

g:Profiler uses passing candidate genes as foreground and all evaluated Phase 1 genes as background. It retains BH-adjusted values.

DAVID addendum

DAVID uses normalized Ensembl identifiers and an uploaded custom background. It counts terms at Benjamini ≤ 0.05.

Filter rule. Do not mix FDR, BH q, EASE, or Benjamini values. These providers test different mapped identifiers, backgrounds, and term sets.

Sources: JOURNEY.md, independent pathway branch. DAVID pathway report.

PATHWAY PIPELINE: PHASE 2 RESULTS

Phase 2 saved enrichment export contains 34 FDR-significant terms



Input and output

Input: Phase 2 candidate seed genes. Output: 84 saved pathway rows with source database, term name, and FDR.

Leading saved terms

Organelle organization, response to stimulus, and cytoskeleton organization have the lowest saved FDR values.

Documentation difference. One narrative report calls all 84 rows significant. A data-blind count of saved FDR values finds 34 rows at FDR ≤ 0.05. This report uses the saved values.

Conclusion. Phase 2 finds broad annotation enrichment. Broad terms do not isolate a unique chromosome-segregation mechanism or show pathway activity.

Data-blind check of allowed pathway-name and FDR fields. Narrative comparison: JOURNEY.md.

Independent g:Profiler and DAVID results remain supplementary

Independent P0–P7

Input. Passing candidate genes against all evaluated Phase 1 genes.

Tool and filter. g:Profiler with provider BH adjustment.

Result. 16,455 returned terms. Zero terms have BH q ≤ 0.05. Lowest q: 0.08037.

Conclusion. No pathway enters primary prioritization.

DAVID addendum

Input. 1,926 submitted foreground and 8,740 submitted background Ensembl identifiers.

Tool and filter. Native DAVID annotation chart. Benjamini ≤ 0.05.

Result. DAVID mapped 1,591 foreground and 8,371 background identifiers. Two of 318 terms pass Benjamini.

Conclusion. Mapping loss changes the effective tested universe. The two term names are unavailable in checked-in documentation.

Provider comparison. 74 stable term identifiers overlap between the saved comparison sets. Shared identifiers describe annotation coverage. They do not prove independent replication.

Sources: JOURNEY.md, P0–P7 results. DAVID pathway report.

PATHWAY PIPELINE: DOWNSTREAM DRUG-SCREENING DECISION

Pathway results do not yet support a separate pathway-driven drug shortlist

Input
Significant pathway terms, matched pathway members, and term-level uncertainty.

→

Required filter
Keep terms with a defensible background, adjusted significance, and a specific directional mechanism.

→

Decision
Only then query pathway-member targets in pharmacology sources and compare them with network-screen targets.

Phase 2 result

34 of 84 saved rows meet FDR ≤ 0.05. Leading terms are broad. They do not provide a direction of pathway change or a target-action rationale.

Independent pathway result

P0–P7 has zero terms at BH q ≤ 0.05. DAVID has two significant terms, but checked-in documentation does not name them.

Downstream result. This deck reports no pathway-driven drug shortlist. P6 therapeutic expansion is a documented method, not a completed pathway-specific drug-screening result.

Next validation. Measure a directional pathway defect in an appropriate cell model. Then test a target and its drug action against a prespecified rescue endpoint.

P0–P7 workflow: JOURNEY.md. Drug-screening decision rules: drug-repurposing context.

CONCLUSION AND NEXT STEPS

Use the pipelines to prioritize follow-up, not to make clinical conclusions

What the combined workflow provides

- Published context for known MVA-related genes and processes.
- A documented candidate seed-gene list from VCF evidence filters.
- A strict, no-expansion STRING network context.
- A separate network-based pharmacology inventory with safety tiers.
- Three pathway-analysis views with explicit statistics and backgrounds.

What still requires validation

- Variant confirmation and allele phase.
- Cellular or functional evidence for a proposed mechanism.
- Reconciliation of code, saved tables, and narrative counts.
- Pathway analysis with a defensible callable-gene background.
- Target engagement, exposure, toxicity, and rescue assays for every drug hypothesis.

Final conclusion. Literature, candidate-gene filtering, network context, pathway analysis, and separate downstream drug decisions form a reproducible research workflow. The outputs generate hypotheses. They do not diagnose disease or demonstrate therapeutic benefit.

Source map

[1] Literature workflow reproduction.

[2] Literature review specification.

[3] End-to-end analytical journey.

[4] Candidate-gate implementation.

[5] DAVID pathway report.

[6] Drug-repurposing evidence and safety context.

[7] Hanks et al., 2004, BUB1B and constitutional aneuploidy.